

# Sam Crouch

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## EDUCATION

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### Missouri University of Science and Technology

Rolla, MO

*B.S. Computer Science*

*May 2027*

**Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Programming Languages, Microcontrollers and Embedded Systems, File Structures and Database Systems, Introduction to Artificial Intelligence

## EXPERIENCE

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### Controls Intern

May 2026 – Present

*Altec*

*Saint Joseph, MO*

- Developed a Qt/QML library for embedded Linux displays, standardizing UI components across multiple product lines
- Implemented a C++ daemon to expose hardware functionality through a text file interface, decoupling application logic from low-level hardware control

### Forklift Operator

May 2025 – Aug. 2025

*Menards*

*Saint Peters, MO*

- Operated forklifts in a lumber yard and warehouse
- Maintained product inventory and order fulfillment

### Lube Tech

Nov. 2022 – Aug. 2024

*Valvoline Instant Oil Change*

*Saint Peters, MO*

- Performed preventative maintenance services including oil changes, fluid replacements, and filter installations
- Conducted vehicle inspections and communicated maintenance recommendations to customers

## PROJECTS

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### pyVPW | *Python*

Sep. 2025 – Present

- Developed an open-source Python library implementing the SAE J1850 VPW protocol for GM vehicle diagnostics
- Designed the API to simplify development of diagnostic and data-logging tools
- Implemented the GM seed-key security algorithm to unlock PCMs, enabling writes to protected memory for tuning and diagnostic applications
- Published the project on GitHub with documentation and example applications, supporting adoption by other developers

### LLM-TUI | *Python, Textual*

Feb. 2026 – May 2026

- Built a customizable, terminal-based AI assistant using Python and the Textual framework
- Designed an asynchronous terminal UI supporting real-time streaming responses, on-the-fly model switching, file attachments, and a snipping-tool style screenshot tool
- Integrated multiple LLM providers including OpenAI, Anthropic, Deepseek, Grok, and Ollama behind a common interface, enabling seamless provider switching

## TECHNICAL SKILLS

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**Languages:** C, C++, Python, QML, Bash, SQL, HTML/CSS

**Frameworks:** Qt, Textual, Django

**Developer Tools:** Git, Docker, CMake, Conan, Linux, GDB

**Embedded/Automotive:** Embedded Linux, CAN Bus, UART, AVR Microcontrollers